

FIN-220 Hands-On Machine Learning

HW4 – Daniel Savage

Suppose you are using polynomial regression. You plot the learning curves, and you notice that there is a large gap between the training error and the validation error.

What is happening?

Most likely the model is overfitting. “the hallmark of overfitting the model”

The model fits the training data well. On the validation set, the model performs poorly. This is data the model hasn’t seen before.

In the context of using polynomial regression, the degree of the model is probably too high. It is essentially fitting noise in the training data.

What are three ways to solve this?

Get more training data.

Clean up the training data.

Reduce Model Complexity.

The book at this point brings up Regularized Linear Models: Ridge Regression, Lasso Regression, and Elastic Net Regression. Then for iterative learning models: Early Stopping. I’m assuming that these topics are subsequent to this question.

2) Suppose you are using ridge regression, and you notice that the training error and the validation error are almost equal and fairly high.

Would you say that the model suffers from high bias or high variance?

High bias. The model is underfitting the data. Because both training and validation error are high, the model does not fit either set well. The problem does not state if we are fitting a linear model or have the option to fit a polynomial model. The problem may benefit from a higher polynomial degree.

Should you increase the regularization hyperparameter or reduce it?

Reduce the regularization hyperparameter.

Reducing the regularization hyperparameter, with a polynomial model will, in effect, also increase the polynomial degree.

3) Suppose you want to classify pictures as outdoor/indoor and daytime/nighttime.

Should you implement two logistic regression classifiers or one softmax regression classifier?

We are going to want two logistic regression classifiers. It is a multi-label classification problem. An image can be outdoor daytime, outdoor nighttime, indoor daytime, or indoor nighttime. But indoor/outdoor and daytime/nighttime are independent of each other.